Conservation Genetics (2022) 23:713–725
https://doi.org/10.1007/s10592-022-01450-w

RESEARCH ARTICLE

Genomic population structure of Grass Pickerel (*Esox americanus vermiculatus*) in Canada: management guidance for an at-risk fish at its northern range limit

Nathan K. Lujan[1,2,3] · Julia E. Colm[4] · Jason T. Weir[1] · Fielding A. Montgomery[1] · Brice P. Noonan[2] · Nathan R. Lovejoy[1] · Nicholas E. Mandrak[1]

Received: 13 September 2021 / Accepted: 6 May 2022 / Published online: 27 May 2022
© Crown 2022

Abstract

Eighty nine (42%) of Canada's 215 freshwater fish species have been assessed as at risk by the Committee on the Status of Endangered Wildlife in Canada. This study examines genomic population structure of the at-risk Grass Pickerel (*Esox americanus vermiculatus*), a small (≤ 33 cm) predatory fish that in Canada has a range spanning approximately 114,000 km^2 of southern Ontario. Within this range it occupies approximately ten sites that are mostly shallow, weedy, and slow-flowing. Its populations and habitat are declining. This study defines population clusters and quantifies genomic diversity within and between populations based on > 5500 loci and > 950 SNPs from genomes of 66 individuals representing the subspecies' entire Canadian range. Ordination and STRUCTURE analyses revealed four major geographic/genomic clusters centered in the Georgian Bay-Severn River, southeastern shore of Lake Huron, Niagara Peninsula, and upper St. Lawrence River. Major clusters were distinguished by relatively high Hudson *Fst* values (0.205–0.480), with Georgian Bay-Severn River being consistently most distinct. The Niagara Peninsula major cluster contained an additional three discernable sub-clusters differentiated by *Fst* values as great or greater than major clusters, despite spanning only ca. 200 km^2. Genomically distinct Niagara sub-clusters occurred in Abino Drain, Big Forks Creek, and Tea Creek. Samples from sites between both major and minor clusters exhibited admixture from adjacent clusters. Despite current management of Grass Pickerel under a single designatable unit throughout its Canadian range, we map considerable geographic population structure that should help guide the designation of additional conservation units.

Keywords Great Lakes · At-risk subspecies · Designatable unit · Population structure · Niagara Peninsula · Postglacial colonization

Introduction

Freshwater ecosystems occupy < 0.1% of the Earth's surface yet host approximately 60% of all described species, with freshwater fishes being among the most imperiled vertebrates globally (Dudgeon et al. 2006; Su et al. 2021). In North America, approximately 39% of fish species are considered imperiled and this list is increasing (Jelks et al. 2008), whereas in Canada, 89 (42%) of Canada's 215 freshwater fish species have been assessed as at risk by the Committee on the Status of Endangered Wildlife in Canada (COSEWIC). Freshwater fishes are threatened in part because they commonly occur at low abundances and exhibit restricted or patchy site occupancy—often due to specific habitat requirements (Sheldon 1988; Magurran and Henderson 2010). Throughout North America, habitat

✉ Nathan K. Lujan
nlujan@rom.on.ca

1 Department of Biological Sciences, University of Toronto Scarborough, Toronto, ON M1C 1A4, Canada

2 Department of Biological Sciences, University of Mississippi, Oxford, MS 38677, USA

3 Present Address: Department of Natural History, Royal Ontario Museum, Toronto, ON M5S 2C6, Canada

4 Fisheries and Oceans Canada, 867 Lakeshore Road, Burlington, ON L7S 1A1, Canada

degradation, invasive species, pollution, and overexploitation pose significant threats to freshwater fishes (Dextrase and Mandrak 2006; Jelks et al. 2008; McCune et al. 2013). Rarity and patchy distributions, combined with limited connectivity across linear stream networks or disjunct lakes, contribute to often high population-level genetic diversity in freshwater fishes (Ward et al. 1994)—one of three forms of biodiversity prioritized for conservation by the International Union for the Conservation of Nature (IUCN; McNeely et al. 1990). Genetic diversity is essential as the template on which evolution acts, providing populations with the evolutionary flexibility to respond to stochastic environmental and anthropogenic stressors (Reed & Frankham 2003).

The Grass Pickerel (*Esox americanus vermiculatus* LeSueur 1846) is a small predatory fish (≤ 33 cm; Fig. 1) that in Canada has a range spanning approximately 114,000 km^2 of southern Ontario and the southwestern corner of Quebec (Crossman and Holm 2005; Fig. 2C). Within this range, it is only known from approximately 10 locations (COSEWIC 2014; Fig. 2C); however, the subspecies is widespread and occasionally abundant throughout the southern Great Lakes, central Mississippi River basin, and Gulf Coastal plain in the United States. Grass Pickerel habitat is typically shallow (< 2 m), heavily vegetated, slow-moving, lowland streams and wetlands, overflow ponds of large streams, and stream expansions with mud or organic substrates and clear- to tea-coloured water that is mildly alkaline to slightly acidic (Crossman 1962; Scott and Crossman 1998; Crossman and Holm 2005; Cain et al. 2008). The sister subspecies to Grass Pickerel is the Redfin Pickerel (*Esox americanus americanus* Gmelin 1789), whose range extends down the Atlantic Coast of the United States and overlaps with Grass Pickerel in parts of the Gulf Coast. Together, these *Esox americanus* subspecies are closely related to the Chain Pickerel (*Esox niger* LeSueur 1818), which has a range largely overlapping that of the Redfin Pickerel. During recent Pleistocene glacial maxima, Grass Pickerel was isolated from Redfin and Chain Pickerel by the Appalachian highlands, but ranges of all three species have expanded northward following glacial retreat and are now coming into secondary contact. Today, the easternmost range of Grass Pickerel in Canada, in the upper St. Lawrence River, overlaps with Redfin and Chain pickerels. Hybridization with these eastern lineages is a specific threat to genetic diversity and conservation of Grass Pickerel in Canada.

Grass Pickerel is listed as Special Concern under the *Species at Risk Act* (SARA) based on its restricted occurrence in few, disjunct locations with small population sizes (COSEWIC 2014), and is managed throughout its Canadian range under a single designatable unit (Beauchamp et al. 2012). Subspecific conservation units, termed designatable units in Canada (COSEWIC 2018), must not only be genetically or geographically discrete but also be 'evolutionarily significant' (COSEWIC 2018). Evolutionary significance can by demonstrated either by identifying genomic regions under positive selection or by measuring functional morphological specializations. It can also be inferred based on deep intraspecific phylogeography (e.g., originating in different Pleistocene refugia) or occurrence in strongly divergent habitats likely to result in heritable local adaptions (COSEWIC 2018).

In addition to hybridization from congeners, Grass Pickerel in Canada is threatened by wetland loss, agricultural drain maintenance, and other human activities. The subspecies is known to avoid areas with significant groundwater inputs or high reach slopes (Colm et al.

Fig. 1 Canadian Grass Pickerel (*Esox americanus vermiculatus*) in **A** lateral view from Walpole Island, ON, on Lake St. Clair, and **B** dorsal view from Twenty-Mile Creek near Hamilton, ON. Photo **A** by FAM, photo **B** by George Coker

2019), and recent research suggests that the subspecies may have more narrow habitat requirements in Canada. This may explain why the subspecies is patchily distributed across few disjunct waterbodies and absent from intervening areas of well-sampled, seemingly suitable habitat (Colm et al. 2019). Moreover, census-based population estimates at three Grass Pickerel locations in Canada have been declining (Beauchamp et al. 2012; Colm 2015), highlighting the urgency of a better understanding of this subspecies' population structure and population-level genetic diversity.

Here, we use a triple-digest RADseq approach (Bayona-Vásquez et al. 2019; Lujan et al. 2020) to sequence thousands of genome-wide loci from 66 Grass Pickerel individuals from throughout the subspecies' Canadian range. In addition to providing a non-lethal means of characterizing population structure and connectivity within species at risk, genomic tools such as RADseq can help to identify the historical and contemporary ecosystem processes influencing population structure (Poissant et al. 2005; Cook et al. 2007; Stepien et al. 2007; Caldera and Bolnick 2008; Boizard et al. 2009; Lujan et al. 2020), colonization pathways leading to contemporary distributions (Mäkinen et al. 2006), the distribution and provenance of introduced populations (Dlugosch and Parker 2008; Beneteau et al. 2012), and isolated populations requiring special conservation attention (Manel et al. 2003; Cook et al. 2007; Storfer et al. 2007). Moreover, the genetic identification of fine-scale demographic patterns, such as dispersal and gene-flow among populations, may be used to predict the likelihood and success of natural or assisted recolonization of extirpated habitat patches (Bohonak 1999; Palsbøll et al. 2006).

The aim of this study is to characterize contemporary population structure and genomic diversity of Grass Pickerel in Canada to help guide discussions around the designation of additional management units. In general, we expect high genetic structure among localities, even at small spatial scales, due to the subspecies' philopatry, avoidance of open waters, such as main stream and river channel habitats, and preference for shallow, heavily vegetated habitat. Additionally, we expect patterns of genetic divergence to correspond to a combination of both geographic distance (i.e., isolation by distance; Wright 1942) and hydrologic discontinuity (i.e., barriers; Manel et al. 2003), although postglacial recolonization from different refugia and from different patterns of historical connectivity have been shown to affect the contemporary genetic structure of several co-distributed fish species and may likewise affect Grass Pickerel (April and Turgeon 2006; Stepien et al. 2007; Ginson et al. 2015; Lujan et al. 2020).

Methods

Grass Pickerel distribution in Canada

Over 14,200 fish sampling events in Ontario dating from 2002 to 2017 were compiled and mapped for this study from the Fisheries and Oceans Canada (DFO) Biodiversity Science database. From this sampling effort, 66 records for which Grass Pickerel tissue samples were available for genomic analysis were selected (Table 1). These records spanned 10 localities within a total area of approximately 114,000 km^2, including lakes and tributaries in the Georgian Bay-Severn River drainage, tributaries of the southeastern shore of Lake Huron, northeastern tributaries of Lake St. Clair, tributaries to Lake Erie at the base of Long Point (including Big Creek National Wildlife Refuge), agricultural drains in the Niagara region, Prince Edward County, and a few localities along the upper St. Lawrence River in eastern Ontario and near the border with Quebec (Fig. 2C).

Tissue collection

Fin clips from the lower caudal lobe were non-lethally collected from 66 individuals between 2009 and 2016 by DFO. Specimens were collected using combinations of seines and fyke nets. All fish were released alive after being clipped. Tissue samples were stored in 95% ethanol. Between one and 23 individuals were sampled from a given site, with sites spanning the range of Grass Pickerel in Canada (Fig. 2).

Restriction-site associated DNA-sequencing (RADseq) library preparation, sequencing, and trimming

Whole genomic DNA was extracted using a standard salt extraction protocol (Lujan et al. 2020) and quantified using a Qubit flourometer (Thermo Fisher Scientific Inc., MA, USA). For each sample, 100 ng of whole genomic DNA was digested and ligated to barcodes and Illumina sequencing adaptors using a three-enzyme protocol developed by Bayona-Vásquez et al. (2019), with modifications by Lujan et al. (2020). The enzymes EcoR1 and XbaI were used to digest the genome and NheI was used to separate adapter-dimers formed by phosphorylated adapters. Libraries were sequenced using an Illumina NextSeq 550 high-throughput benchtop sequencer (Illumina Inc., CA, USA) using the NextSeq 500/550 High Output 75 cycle kit. Reads were demultiplexed using the Illumina bcl2fastq conversion software programmed to allow for up to two barcode mismatches (–barcode-mismatches) and trimmed to a standard length of 56 bp using the program FASTQ/A Trimmer

Table 1 Locality of origin for 66 tissues of Grass Pickerel (*Esox americanus vermiculatus*) examined in this study

Sample	Province	Regional cluster	Locality	Latitude	Longitude
2921	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3273	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3274	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.74694	– 79.39362
3275	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3283	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3284	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3285	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3286	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3426	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3427	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3428	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3429	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3430	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3431	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3432	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3433	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3434	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3435	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3436	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3438	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3439	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3440	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
3441	Ontario	Georgian Bay-Severn River	Kahshe Lake	44.87744	– 79.2136
2473	Ontario	Lake St. Clair	Little Bear Creek	42.52566	– 82.27116
2474	Ontario	Lake St. Clair	Little Bear Creek	42.52566	– 82.27116
3372	Ontario	Lake St. Clair	Little Bear Creek	42.51703	– 82.37724
2410	Ontario	Long Point	Long Point Bay	42.58445	– 80.41723
T10828	Ontario	Long Point	Hahn Marsh	42.58229	– 80.51985
577	Ontario	Niagara	Big Forks Creek	42.95108	– 79.36827
579	Ontario	Niagara	Tea Creek	43.03042	– 79.10624
580	Ontario	Niagara	Tea Creek	43.03708	– 79.08839
583	Ontario	Niagara	Tea Creek	43.03127	– 79.10583
584	Ontario	Niagara	Tea Creek	43.03127	– 79.10583
672	Ontario	Niagara	Lyon's Creek	42.98429	– 79.20025
673	Ontario	Niagara	Lyon's Creek	42.99655	– 79.16404
675	Ontario	Niagara	Lyon's Creek	42.99655	– 79.16404
2985	Ontario	Niagara	Abino Drain	42.93420	– 79.07671
3144	Ontario	Niagara	Abino Drain	42.93420	– 79.07671
3453	Ontario	Niagara	Abino Drain	42.91435	– 79.12917
3459	Ontario	Niagara	Big Forks Creek	42.94925	– 79.38892
3463	Ontario	Niagara	Abino Drain	42.91435	– 79.12917
3466	Ontario	Niagara	Abino Drain	42.92807	– 79.10788
3510	Ontario	Niagara	Big Forks Creek	42.94785	– 79.34784
3512	Ontario	Niagara	Big Forks Creek	42.94785	– 79.34784
3513	Ontario	Niagara	Big Forks Creek	42.94920	– 79.38881
3514	Ontario	Niagara	Big Forks Creek	42.94920	– 79.38881
3522	Ontario	Niagara	Big Forks Creek	42.94913	– 79.38825
3523	Ontario	Niagara	Big Forks Creek	42.94913	– 79.38825
3527	Ontario	Niagara	Abino Drain	42.92807	– 79.10788
T05881	Ontario	Niagara	Tea Creek	43.00694	– 79.11250
T05882	Ontario	Niagara	Tea Creek	43.00694	– 79.11250

Table 1 (continued)

Sample	Province	Regional cluster	Locality	Latitude	Longitude
587	Ontario	Prince Edward County	West Lake	43.93953	– 77.25453
2510	Ontario	Saint Lawrence	Jones Creek	44.50103	– 75.87828
4053	Quebec	Saint Lawrence	Rivière aux Saumons	45.02304	– 74.52013
4055	Quebec	Saint Lawrence	Chenaux Lac-Saint-Francois	45.06015	– 74.44085
4059	Quebec	Saint Lawrence	Ruisseau McMillan	45.05547	– 74.44775
4060	Quebec	Saint Lawrence	Ruisseau McMillan	45.05633	– 74.44822
4061	Quebec	Saint Lawrence	Ruisseau sans nom	45.07853	– 74.42461
4063	Quebec	Saint Lawrence	Rivière aux Saumons	45.00815	– 74.51828
616	Ontario	South Lake Huron	Old Ausable Channel	43.26487	– 81.81627
617	Ontario	South Lake Huron	Old Ausable Channel	43.27175	– 81.80450
678	Ontario	South Lake Huron	L Lake	43.22537	– 81.90773
679	Ontario	South Lake Huron	L Lake	43.22466	– 81.91744
682	Ontario	South Lake Huron	L Lake	43.22460	– 81.90664
683	Ontario	South Lake Huron	L Lake	43.22472	– 81.91307
T05903	Ontario	Twenty Mile Creek	Twenty Mile Creek	43.15056	– 79.83417

within the FASTX-Toolkit (Gordon & Hannon 2010). All other program settings were left at default. See Lujan et al. (2020) for details.

Read clustering and SNP calling

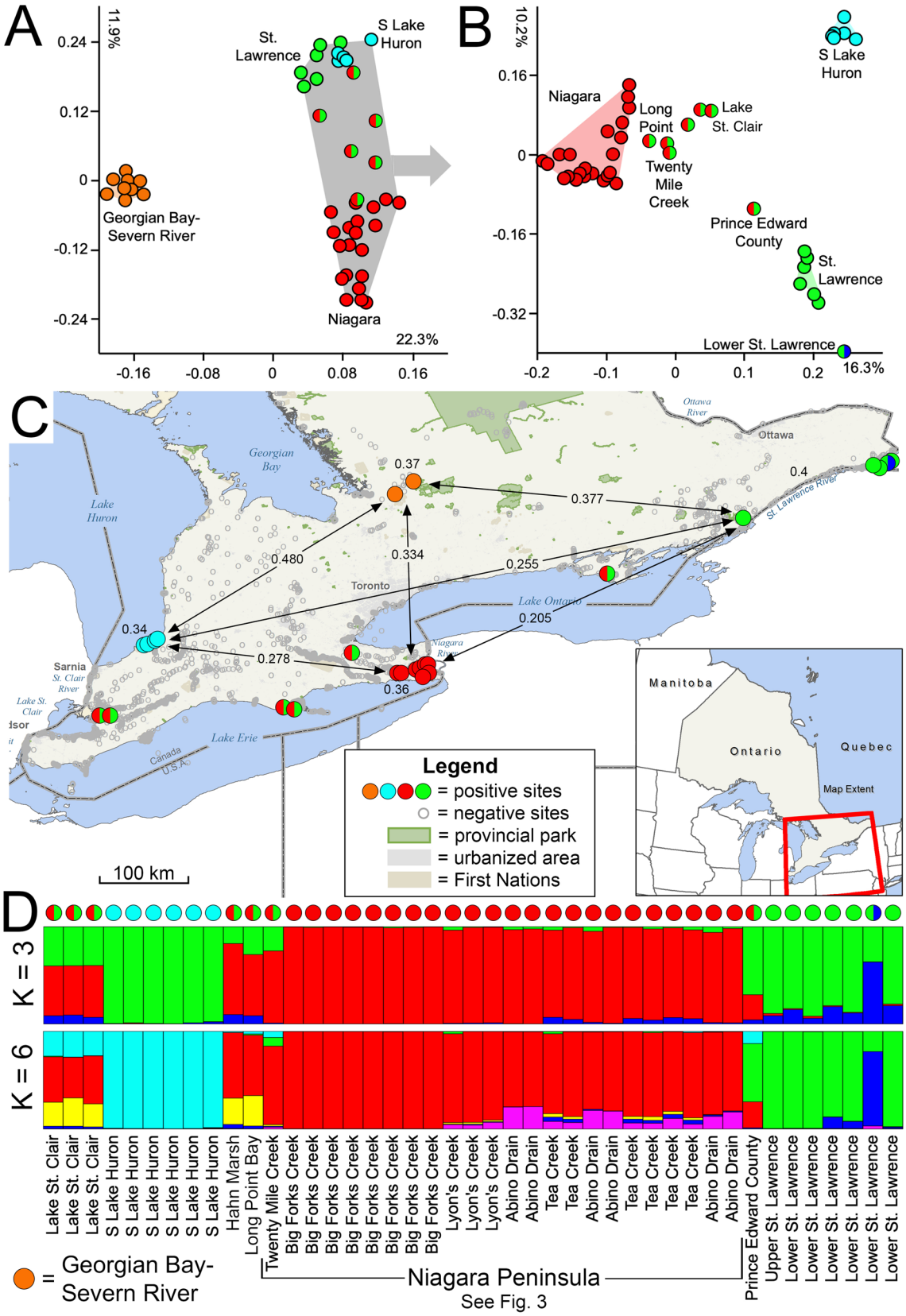
The program pyRAD v3 (Eaton 2014) was used to cluster trimmed reads both within and across samples and to identify single nucleotide polymorphisms (SNPs). To maximize processing efficiency, demultiplexed reads for each sample were run through pyRAD Steps 2–4 (filtering, within-sample clustering, and error rate and heterozygosity estimates) on separate processors allocated 3 GB of memory each. Results of sample-specific within-group clustering were combined for pyRAD steps 5–7 (consensus sequence creation, across-sample clustering, alignment, paralog filtering and output generation), which were run on a single processor allocated 6 GB of memory. The following parameter values were used for all pyRAD analyses: minimum coverage for a cluster = 10, maximum number of sites with Phred quality score < 20 = 5, clustering threshold = 90%, maximum number of individuals with a shared heterozygous site = 75%, maximum number of SNPs per locus = 1 (latter parameters set stringently to exclude potential paralogous loci).

SNP calling was conducted in six iterations of pyRAD steps 5–7 that differed in sample composition based on results of the STRUCTURE analyses described below. The first iteration included all samples; the second iteration excluded only Georgian Bay-Severn River samples, and iterations 3–6 separately examined each of four subpopulations consistently identified in previous iterations: Georgian Bay-Severn River, Niagara Peninsula, South Lake Huron, St. Lawrence River. Iteration 1 (all samples) was programmed to retain only loci present in at least 50 of 66 individuals (76%); iteration 2 (all except the Georgian Bay-Severn River) retained only loci present in at least 40 of 43 individuals (93%), and iterations 3–6 retained loci present in at least 17 of 23 (74%) of Georgian Bay-Severn River population individuals, 17 of 23 (74%) of Niagara Peninsula population individuals, 4 of 6 (67%) South Lake Huron individuals, or 4 of 6 (67%) St. Lawrence River individuals.

Determining population genomic structure and dissimilarity

Population genomic structure was examined using three approaches. The first examined genomic diversity unaffected by assumptions of population structure. For this, we conducted a principal coordinates analysis (PCoA) of 'unlinked' SNPs ('unlinked' being what pyRAD calls SNPs that were sampled as a single SNP per locus) using the PCoA function in the program PAST v3 (Hammer et al. 2001). Although estimates of population structure and genetic diversity based on RADseq datasets such as ours can potentially be biased by the failure to sequence some alleles from some individuals or populations due to mutations in restriction cut sites (known as allele dropout), theoretical modelling finds such biases to be low (Gautier et al. 2013).

The second approach examined how much geographic structure and genetic admixture is apparent when SNP data are fit to an arbitrary range of subdivisions or 'populations' (K). For this, we used the program STRUCTURE (Pritchard et al. 2000), which was parameterized to run three iterations of each K-value, with each iteration running for 100,000 burn-in and 66,000 recorded generations. Using output from the most relevant pyRAD iteration (see above), multiple iterations of STRUCTURE analyses were run on: (1) all populations and samples (evaluating K-range: 1–8); (2)


A
11.9%
0.24
0.12
0
-0.12
-0.24
St. Lawrence
S Lake Huron
Georgian Bay-Severn River
Niagara
22.3%
-0.16
-0.08
0
0.08
0.16
B
10.2%
0.16
0
-0.16
-0.32
S Lake Huron
Niagara
Long Point
Lake St. Clair
Twenty Mile Creek
Prince Edward County
St. Lawrence
Lower St. Lawrence
16.3%
-0.2
-0.1
0
0.1
0.2
C
Lake Huron
Georgian Bay
Ottawa River
Ottawa
St. Lawrence River
Toronto
Lake Ontario
Niagara River
Sarnia
St. Clair River
Lake St. Clair
Lake Erie
Canada
U.S.A.
0.37
0.377
0.4
0.480
0.334
0.255
0.205
0.34
0.278
0.36
100 km
Legend
= positive sites
= negative sites
= provincial park
= urbanized area
= First Nations
Manitoba
Ontario
Quebec
Map Extent
D
K = 3
K = 6
Lake St. Clair
Lake St. Clair
Lake St. Clair
S Lake Huron
S Lake Huron
S Lake Huron
S Lake Huron
S Lake Huron
S Lake Huron
Hahn Marsh
Long Point Bay
Twenty Mile Creek
Big Forks Creek
Big Forks Creek
Big Forks Creek
Big Forks Creek
Big Forks Creek
Big Forks Creek
Big Forks Creek
Big Forks Creek
Lyon's Creek
Lyon's Creek
Lyon's Creek
Abino Drain
Abino Drain
Tea Creek
Tea Creek
Abino Drain
Abino Drain
Tea Creek
Tea Creek
Tea Creek
Tea Creek
Abino Drain
Abino Drain
Prince Edward County
Upper St. Lawrence
Lower St. Lawrence
Lower St. Lawrence
Lower St. Lawrence
Lower St. Lawrence
Lower St. Lawrence
Lower St. Lawrence
Lower St. Lawrence
= Georgian Bay-Severn River
Niagara Peninsula
See Fig. 3

◂**Fig. 2** Genomic population structure of Grass Pickerel (*Esox americanus vermiculatus*) in Canada described by principal coordinates (**A**, **B**) and STRUCTURE analyses (C, D) of 66 individuals and 740–950 SNPs per individual. Given the considerable genomic distinctiveness of samples from the Georgian Bay-Severn River (**A**), and the lack of apparent admixture from this cluster to others (STRUCTURE results not shown), these samples were excluded from subsequent PCoA (**B**) and STRUCTURE analyses (**D**). In map **C**, mean pairwise Hudson *Fst* values are given on arrows between clusters. See Table 2 for *Fst* estimate confidence intervals. STRUCTURE results presented as barplot (**D**) in which each column is an individual and each color is a distinct genomic cluster, with percentage of an individual's genome assignable to a cluster shown by color proportion within columns. Analysis of Evanno's delta K yielded K = 3 and K = 6 as optimal subdivisions of genomic diversity (see Fig. S1). Percent variation explained by each PCoA axis (**A**, **B**) given on axis. Icon colors in PCoA scatterplot (**A**, **B**) derived from STRUCTURE analysis. Map **C** derived from > 14,200 fish sampling events dating from 2002 to 2017 in the Fisheries and Oceans Canada (DFO) Biodiversity Science database

all populations and samples except members of the highly distinct Georgian Bay-Severn River population (evaluating K-range: 1–8); and, (3) separate analyses of each of the four major initial population clusters (Georgian Bay-Severn River, Niagara Peninsula, South Lake Huron, St. Lawrence River; evaluating K-range: 1–4 each). To determine the optimal number of population subdivisions, results of the STRUCTURE analysis were uploaded to the online Structure Harvester website (Earl and von Holdt 2012) to generate a distribution for Evanno's delta K (Evanno et al. 2005; Fig. S1) from which peaks were interpreted as optimal numbers of population clusters (K).

For the third approach, we calculated Hudson's F_{st} following the equation provided in Bhatia et al. (2013) using an R script from Barrera-Guzman et al. (2018). Bhatia et al. (2013) found that historically commonly used fixation indices, such as Nei (1973) and Weir and Cockerham (1984), which calculate the average of ratios for individual markers, are sensitive to sample size differences and can result in large reductions in F_{st} estimates for genomic data. Thus, Bhatia et al. (2013) recommended using Hudson's F_{st}, which takes the ratio of an average calculated for all markers. Significance of each F_{st} estimate was determined by calculating 95% confidence intervals from a 1000 replicate bootstrap using the R script from Barrera-Guzman et al. (2018). Pairwise *Fst* estimates were calculated using only individuals with ≥ 88% assignment to a given genomic cluster to minimize potential bias from heavily admixed individuals.

Patterns of population structure and dissimilarity within the Niagara Peninsula

STRUCTURE and Structure Harvester analysis of only the Niagara Peninsula populations revealed geographically concordant population structure, motivating a more restricted analysis of this region using the same three approaches to mapping population structure as described above. See Fig. S2 for the Evanno's delta K distribution (Evanno et al. 2005) from this analysis.

Quantifying genomic diversity

Observed (*Ho*) and expected (*He*) heterozygosities were calculated for each locus and each cluster using the divBasic function in the R package diveRsity (Keenan et al. 2013). To minimize potential bias from admixture, heterozygosities were calculated using only individuals with ≥ 88% assignment to a given genomic cluster.

Results

Sequencing results

A total of 377,828,162 demultiplexed reads were obtained for an average of 4,951,276 reads per sample (range 629,730–29,435,347). The pyRAD analysis of the entire data set yielded 5594 loci shared by at least 50 of 66 individuals (76%) and 956 'unlinked' bi-allelic SNPs ('unlinked' being pyRAD's description of single SNPs from different sequenced loci, though no specific test for linkage was performed). Analysis of all samples except those from Georgian Bay-Severn River yielded 4755 loci and 742 'unlinked' bi-allelic SNPs. Population-specific pyRAD analyses yielded the following: Georgian Bay-Severn River: 6122 loci, 143 SNPs; Niagara Peninsula: 16,736 loci, 1295 SNPs; South Lake Huron: 16,300 loci, 510 SNPs; St. Lawrence River: 13,997 loci, 626 SNPs.

Large-scale patterns of genomic structure

Principal coordinates and STRUCTURE analyses of samples from throughout the range of Grass Pickerel in Canada yielded highly congruent results in which four distinct geographic and genomic clusters were identified (Fig. 2). Individuals from sites between these clusters exhibited admixture from neighboring clusters in approximate proportion to their geographic proximity. All three approaches to quantifying and visualizing genomic structure (PCoA, STRUCTURE, F_{st}) identified individuals from Georgian Bay-Severn River as the most genetically distinct (Fig. 2A, C; STRUCTURE results not shown). Given the degree to which Georgian Bay-Severn River samples were genetically distinct from all others (Fig. 2A), the initial PCoA and STRUCTURE analyses of all samples yielded consistent support for only two population clusters (K = 2) separating Georgian Bay-Severn River individuals from all others. Thus, we present PCoA results with (Fig. 2A) and without (Fig. 2B) Georgian Bay-Severn River samples, but

only STRUCTURE results for analyses that excluded Georgian Bay-Severn River samples (Fig. 2D).

The PCoA analysis of non-Georgian Bay-Severn River samples (Fig. 2B) distributed individuals across an approximately equilateral triangle, with individuals from the following localities clustered at the vertices: Niagara, South Lake Huron, and St. Lawrence River. Individuals from sites between these clusters, such as Lake St. Clair, Long Point, and Prince Edward County occupied positions within the triangle in general concordance with their geographic proximity to the three main clusters. The only individual occurring outside of this triangle was from a St. Lawrence River site near the Ontario-Quebec border that is potentially exposed to introgression from closely related Redfin and Chain Pickerel (Fig. 2B).

STRUCTURE analyses of non-Georgian Bay-Severn River samples yielded similar results, but with Evanno's delta K peaks occurring at both K = 3 and K = 6. The K = 3 results identified only two geographic clusters, one exclusively from the Niagara region and another that combined samples from South Lake Huron and from ~ 450 km east in the St. Lawrence River, with both clusters being geographically closer to the Niagara cluster than they are to each other (Fig. 2C, D). However, the K = 6 results recognized the South Lake Huron and St. Lawrence River samples as distinct clusters. Consistent with the PCoA results, individuals from localities between these population clusters exhibited patterns of admixture from adjacent population clusters. Notably, the same Lower St. Lawrence River individual that was outside of the PCoA triangle in Fig. 2B also exhibited significant (> 50%) admixture from an otherwise largely unobserved genomic cluster (Fig. 2D).

Large-scale patterns of genomic dissimilarity and genomic diversity

Pairwise comparisons of the four population clusters delineated by PCoA and STRUCTURE analyses yielded mean *Fst* estimates ranging from 0.205 (Niagara-St. Lawrence) to 0.480 (Georgian Bay-Severn River-South Lake Huron) (Fig. 2; Table 2). Consistent with the PCoA and STRUCTURE analyses, the Georgian Bay-Severn River population was consistently the most genetically dissimilar to other populations, with mean *Fst* estimate ranging from 0.344 (vs. Niagara) to 0.480 (vs. South Lake Huron). Ninety-five percent confidence intervals indicated that all *Fst* estimates were significantly greater than zero (Table 2). The minimum *Fst* value of the 95% confidence intervals was 0.164 for Niagara-St. Lawrence (Table 2). A Mantel test of correlation between the geographic coordinates (latitude vs. longitude) and the genomic principal coordinates (PC1 vs. PC2) of all samples found no correlation between geographic and genetic distance.

All four population clusters exhibited similar heterozygosity values, ranging from 0.34 (South Lake Huron) to 0.4 (St. Lawrence River; Fig. 2C). Observed heterozygosities (*Ho*) for all populations were significantly greater than expected heterozygosities (*He*) given Hardy–Weinberg assumptions.

Niagara Peninsula patterns of genomic structure, dissimilarity, and diversity

A STRUCTURE analysis of 23 samples restricted to approximately 200 km^2 of the Niagara Peninsula (Fig. 2B) plus three adjacent samples from near Hamilton (Twenty Mile Creek) and Long Point (Hahn Marsh and Long Point Bay) revealed three geographically concordant genetic sub-clusters within the Niagara Peninsula (including Twenty Mile Creek) and an entirely distinct genetic cluster for Long Point (K = 4; Fig. 3C). Distinct genetic sub-clusters within the Niagara Peninsula were associated with Big Forks Creek, Tea Creek, and Abino Drain. A genomic principal coordinates analysis of these same samples yielded similar results, with samples from Big Forks Creek, Tea Creek, and Abino Drain being clustered at opposite vertices of an approximately equilateral triangle (Fig. 3A). Both the STRUCTU RE analysis and the PCoA yielded evidence for admixture at two sites: (1) at sites in Lyons Creek between Big Forks Creek and Tea Creek clusters, and (2) between Tea Creek and Abino Drain clusters at downstream (hydrologically intermediate) sites in Abino Drain. Within this region, mean pairwise *Fst* estimates were lowest between Abino Drain and Tea Creek (0.143) and highest between Abino Drain and Big Forks Creek (0.265). Genetic diversity (heterozygosity)

Table 2 Pairwise Hudson *Fst* estimates (mean and 95% confidence interval) between the four major Canadian population clusters of Grass Pickerel (*Esox americanus vermiculatus*) identified in this study

	1-Georgian Bay-Severn River	2-Niagara Peninsula	3-South Lake Huron	4-St. Lawrence
1	NA			
2	0.334 (0.285–0.384)	NA		
3	0.480 (0.403–0.533)	0.278 (0.225–0.323)	NA	
4	0.377 (0.313–0.432)	0.205 (0.164–0.250)	0.255 (0.207–0.308)	NA

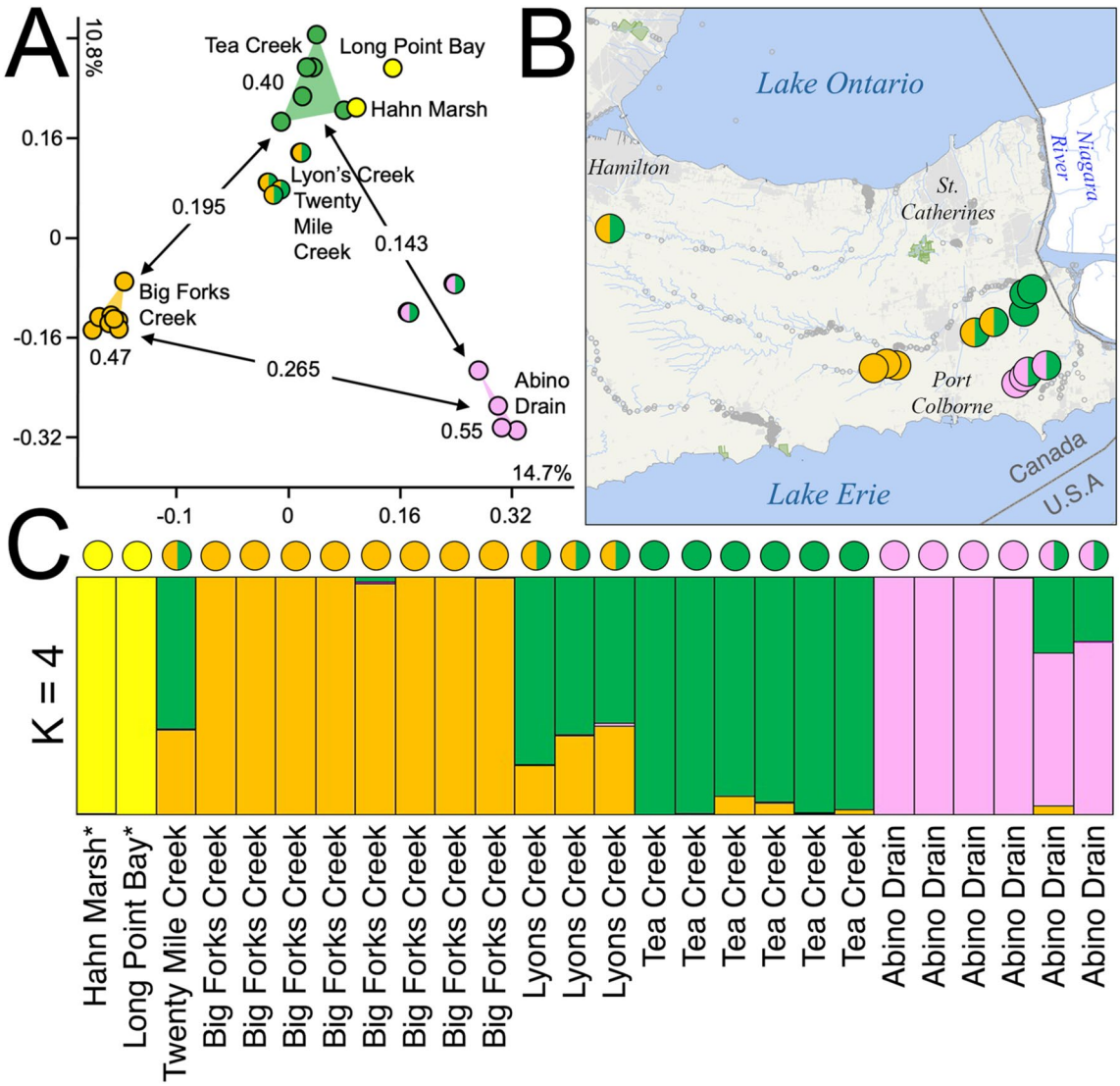


Fig. 3 Genomic population structure of Grass Pickerel (*Esox americanus vermiculatus*) in the Niagara Peninsula described by principal coordinates (**A**) and STRUCTURE (**B**, **C**) analyses of 26 individuals and over 1,690 SNPs per individual. Mean pairwise Hudson *Fst* values given on arrows between PCoA clusters (**A**). See Table 3 for *Fst* estimate confidence intervals. Percent variation explained by each PCoA axis (**A**) given on axis. Icon colors in **A** derived entirely from STRUCTURE results. STRUCTURE results presented as barplot (**C**) in which each column is an individual and each color is a distinct genomic cluster, with percentage of an individual's genome assignable to a cluster shown by color proportion within columns. Analysis of Evanno's delta K yielded K = 4 as an optimal number of subdivisions of genomic diversity. Sample sites Hahn Marsh and Long Point Bay (denoted with asterisk) are outside map boundaries (**B**) but are shown in Fig. 2C. Map **B** derived from fish sampling events dating from 2002 to 2017 in the Fisheries and Oceans Canada (DFO) Biodiversity Science database. Small, open circles represent sampling events that did not yield Grass Pickerel

Table 3 Pairwise Hudson *Fst* estimates (mean and 95% confidence interval) between the three major Niagara Peninsula population clusters of Grass Pickerel (*Esox americanus vermiculatus*) identified in this study

	1-Abino Drain	2-Big Forks Creek	3-Tea Creek
1	NA		
2	0.265 (0.214–0.310)	NA	
3	0.143 (0.108–0.180)	0.195 (0.155–0.244)	NA

within these sub-clusters ranged from 0.40 (Tea Creek) to 0.55 (Abino Drain).

Discussion

Our results provide a detailed picture of genomic structure throughout the Canadian range of Grass Pickerel (*Esox americanus vermiculatus*), filling a critical data gap needed to effectively conserve genetic diversity of this at-risk

subspecies. Results also yield intriguing, but preliminary, insights into the historical biogeographical corridors and processes that may have facilitated the subspecies' northernmost dispersal following glacial retreat. Some of the biogeographical patterns observed in this study are similar to patterns observed in other Great Lakes species, but others are novel and provide insights into the varying hydrological connections that formed across southern Ontario before consolidation of the modern Laurentian Great Lakes. We explore the conservation implications of this study, the anthropogenic and ecological processes that affect contemporary Grass Pickerel population structure in Canada, and the historical processes that likely helped shape its modern distribution.

Conservation implications and recent ecological and anthropogenic processes affecting Grass Pickerel population structure

We demonstrate that Grass Pickerel in Canada comprises at least four genomically distinct, geographically disjunct populations found in the Niagara Peninsula, Georgian Bay-Severn River drainage, southern tributaries of Lake Huron, and tributaries of the upper St. Lawrence River (Fig. 2C). Although our results clearly indicate multiple discrete units, we examined only anonymous genomic markers, the species likely originated in a single Pleistocene refugium (Mandrak and Crossman 1992), and the subspecies occupies largely stereotypical habitats throughout its range in Canada (COSEWIC 2014; Colm et al. 2019). Thus, an annotated genome and functional genomic or similar study is required to further test for additional designatable units in Canada. The phylogenetically closest species having a whole genome sequence available is the Northern Pike (*Esox lucius*), although this genome is not fully annotated (Rondeau et al. 2014). An annotated genome would improve results of demographic studies such as this by allowing the detection and removal of linked loci and would help to identify potentially functional outlier loci under differential selection across populations (Sunde et al. 2022).

Various factors may make Grass Pickerel especially susceptible to impacts on genetic structure (e.g., drift, inbreeding depression, loss of heterozygosity, bottlenecks, founder effects). For example, our results from the Niagara Peninsula (Fig. 3) demonstrate that Grass Pickerel populations in Canada are largely isolated from one another, even when they are geographically close (< 30 river km apart). The subspecies is generally philopatric, with migrations and long-distance dispersals being rare (Crossman 1962; Kramski 2014), and it is a micro-habitat specialist for which unsuitable habitat patches likely act as barriers (Colm et al. 2019). Thus, various human activities, such as construction, channelization, and drain maintenance, may threaten the integrity or persistence of connections within and between Grass Pickerel habitats and should be avoided. Stochastic environmental events present additional potentially negative impacts to population integrity. Examples include alterations of thermal and hydrological processes and increases in the frequency and intensity of extreme weather events due to climate change. Such events can cause mass mortality, reduced spawning and recruitment success, and reductions in summer and winter refugia (Casselman 1996; Doka et al. 2006; Colm 2015; Brinker et al. 2018; Colm et al. 2020).

Historical biogeographical processes giving rise to contemporary Grass Pickerel population structure

In Canada, Grass Pickerel is at the northern limit of a range that extends throughout lowlands of the Great Lakes and Mississippi River basins to the Gulf Coast of Louisiana and East Texas (Page and Burr 2011). Thus, a comprehensive analysis of the subspecies' postglacial dispersal into and throughout the Great Lakes, and in relation to its sister subspecies, Redfin Pickerel, would require a much more geographically broad sample set than examined here. Nonetheless, patterns of population structure and dissimilarity demonstrated in this study exhibit striking concordance with paleo-drainage patterns that have been previously inferred from geological evidence.

All Canadian populations of Grass Pickerel likely dispersed from the Mississippian refugium into the Erie basin through the Fort Wayne Outlet, ca. 14,000 ybp (Mandrak and Crossman 1992). The subspecies subsequently dispersed northward through the St. Clair basin into the Huron basin, and eastward through the Erie basin into the Ontario and upper St. Lawrence basins through glacial outlets and potentially headwater capture. In the Huron basin, the Georgian Bay-Severn River population would have had to disperse into the area during the main phase of glacial Lake Algonquin, ca. 11,000 ypb (Eschman and Karrow 1985), before the lake level dropped and prevented further dispersal into this area from the Huron basin. Although the Georgian Bay-Severn River, southern Lake Huron and Lake St. Clair populations were likely along the same postglacial dispersal pathway, gene flow likely continued between southern Lake Huron and Lake St. Clair populations following the isolation of the Georgian Bay-Severn River population. Similarly, gene flow likely continued among the Lake St. Clair and coastal Lake Erie populations. In the Erie basin, inland populations in the eastern basin of Lake Erie, including Niagara River tributaries, differ from the coastal Lake Erie populations, and exhibit some differences likely due to isolation in the headwaters of local drainage basins. Interestingly, the only population in the western basin of Lake Ontario, Twenty-Mile Creek, occurs on top of the Niagara escarpment and is

separated from Lake Ontario by an 11 m waterfall, which would prevent colonization from the Ontario basin downstream. The population is most similar to the Lyons and Tea Creek populations in the Welland River watershed of the Lake Erie basin, consistent with an hypothesis of headwater capture between the adjacent tributaries of the Welland River in the Erie basin and Twenty-Mile Creek in the Ontario basin (Fig. 3C). Big Forks Creek is a tributary to the upper Welland River and Lyons Creek, and its tributary Tea Creek, are both tributary to the lower Welland River. Construction of the most recent iteration of the Welland Canal in the 1950s cut off flows from the headwaters of Lyons Creek, which are now augmented by water pumped out of the Welland Canal (Yagi and Blott 2012). Siphons were constructed to allow water in the Welland River to flow under the Welland Canal and are a barrier to most fishes (Yagi and Blott 2012). Historical connections, including potential headwater captures between Lyons Creek headwaters and the upper Welland River drainage, may explain the closer relationship of the population in Big Forks Creek to Lyons Creek, than to the closer as the water currently flows Tea Creek. Populations in the eastern Lake Ontario basin (Prince Edward) and upper St. Lawrence are similar, reflecting common colonization patterns. Differences between the upper and lower St. Lawrence populations may be the result of introgression with its sister subspecies (Redfin Pickerel) at the western edge of its range (April et al. 2013).

Conclusions

Despite current management of Grass Pickerel as a single designatable unit throughout its Canadian range, we demonstrate considerable small and large-scale genomic population structure that should inform conservation actions. To determine if additional designatable units are warranted for Grass Pickerel in Canada, the evolutionary significance of populations should be investigated further using annotated genome and functional genomic or similar studies.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10592-022-01450-w.

Acknowledgements We thank Fisheries and Oceans Canada for supporting this research financially and by providing tissue samples, and the Natural Sciences and Engineering Research Council of Canada for funding (NSERC Discovery Grants: RGPIN-2014-05226 to NEM, RGPIN-2016-06221 to NRL, RGPIN-2016-06538 to JTW; NSERC Discovery Accelerator Grant 492890 to JTW). Computations were performed on the Niagara supercomputer at the SciNet High Performance Computing Consortium, funded by the University of Toronto, the Canadian Foundation for Innovation, the Government of Ontario, and Ontario Research Fund – Research Excellence.

Author contributions NKL, NEM, and NRL conceptualized the study, JEC collected samples, NKL and BPN collected data, NKL and JTW analyzed data, FAM prepared maps, NKL, JEC, and NEM drafted the manuscript with input from all authors.

Funding Research was supported by Fisheries and Oceans Canada, the Natural Sciences and Engineering Research Council of Canada (NSERC Discovery Grants: RGPIN-2014-05226 to NEM, RGPIN-2016-06221 to NRL, RGPIN-2016-06538 to JTW; NSERC Discovery Accelerator Grant 492890 to JTW), the University of Toronto, the Canadian Foundation for Innovation, the Government of Ontario, and Ontario Research Fund – Research Excellence.

Data availability All data are made available as supplementary information.

Code availability Code is available from the authors.

Declarations

Conflict of interest We declare no conflicts of interest.

Consent to participate Not applicable.

Consent for publication Not applicable.

Ethical approval Collections, handling, and subsampling of fish for this study was approved by the Fisheries and Oceans Canada institutional review board.

References

April J, Turgeon J (2006) Phylogeography of the banded killifish (*Fundulus diaphanus*): glacial races and secondary contact. J Fish Biol 69:212–228

April J, Hanner RH, Dion-Côte A-M, Bernatchez L (2013) Glacial cycles as an allopatric speciation pump in north-eastern American freshwater fishes. Mol Ecol 22:409–422

Barrera-Guzmán AO, Aleixo A, Shawkey MD, Weir JT (2018) Hybrid speciation leads to novel male secondary sexual ornamentation of an Amazonian bird. Proc Natl Acad Sci 115:E218–E225

Bayona-Vásquez NJ, Glenn TC, Kieran TJ, Pierson TW, Hoffberg SL, Scott PA, Bentley KE, Finger Jr, JW, Watson PR, Louha S, Troendle N, Díaz-Jaimes P, Mauricio R, Faircloth BC (2019) Adapterama III: quadruple-indexed, double/triple-enzyme RADseq libraries (2RAD/3RAD). PeerJ 7.

Beauchamp J, Boyko AL, Dunn S, Hardy D, Jarvis PL, Staton SK (2012). Management plan for the Grass Pickerel (*Esox americanus vermiculatus*) in Canada. Fisheries Oceans Canada Ottawa

Beneteau CL, Walter RP, Mandrak NE, Heath DD (2012) Range expansion by invasion: genetic characterization of invasion of the greenside darter (*Etheostoma blennioides*) at the northern edge of its distribution. Biol Invasions 14:191–201

Bhatia G, Patterson N, Sankararaman S, Price AL (2013) Estimating and interpreting F_{st}: the impact of rare variants. Genome Res 23:1514–1521

Bohonak AJ (1999) Dispersal, gene flow, and population structure. Q Rev Biol 74:21–45

Boizard J, Magnan P, Angers B (2009) Effects of dynamic landscape elements on fish dispersal: the example of creek chub (*Semotilus atromaculatus*). Mol Ecol 18:430–441

Brinker A, Chucholl C, Behrmann-Godel J, Matzinger M, Basin T, Baer J (2018) River damming drives population fragmentation and habitat loss of the threatened Danube streber (*Zingel streber*): implications for conservation. Aquat Conserv Mar Freshwat Ecosyst 28:587–599

Cain ML, Lauer TE, Lau JK (2008) Habitat use of grass pickerel *Esox americanus vermiculatus* in Indiana Streams. Am Midl Nat 160:96–109

Caldera EJ, Bolnick DI (2008) Effects of colonization history and landscape structure on genetic variation within and among threespine stickleback (*Gasterosteus aculeatus*) populations in a single watershed. Evol Ecol Res 10:575–598

Casselman JM (1996) Age, growth and environmental requirements of pike. In: Pike, pp 69–101. Springer, Dordrecht.

Colm JE (2015) Ecology of the imperiled grass pickerel (*Esox americanus vermiculatus* Lesueur) in Ontario: distribution patterns and population decline. Master of Science. Queen's University, Kingston

Colm JE, Mandrak NE, Tufts BL (2019) Local habitat features explain the distribution of the imperiled grass pickerel (*Esox americanus vermiculatus*). Can J Zool 97:479–487

Colm JE, Casselman JM, Mandrak NE (2020) Age, growth, and population assessment of grass pickerel (*Esox americanus vermiculatus*) in two northern populations. Can J Zool 98:527–539

Cook BD, Bunn SE, Hughes JM (2007) Molecular genetic and stable isotope signatures reveal complementary patterns of population connectivity in the regionally vulnerable southern pygmy perch (*Nannoperca australis*). Biol Conserv 138:60–72

COSEWIC (2014) COSEWIC Status appraisal summary for Grass Pickerel (*Esox americanus vermiculatus*). Committee on the Status of Endangered Wildlife in Canada (COSEWIC). Ottawa, ON.

COSEWIC (2018) COSEWIC guidelines for recognizing designatable units. Committee on the Status of Endangered Wildlife in Canada (COSEWIC). Ottawa, ON.

Crossman EJ (1962) The grass pickerel *Esox americanus vermiculatus* LeSueur in Canada. Royal Ontario Museum, Life Sciences Division 55:1–29

Crossman EJ, Holm E (2005) COSEWIC status report on the grass pickerel *Esox americanus vermiculatus* in Canada. Committee on the Status of Endangered Wildlife in Canada, Ottawa, Canada

Dextrase A, Mandrak NE (2006) Impacts of alien invasive species on freshwater fauna at risk in Canada. Biol Invasions 8:13–24

Dlugosch KM, Parker IM (2008) Founding events in species invasions: genetic variation, adaptive evolution, and the role of multiple introductions. Mol Ecol 17:431–449

Doka S, Bakelaar C, Bouvier LD (2006) Chapter 6. Coastal wetland fish community assessment of climate change in the lower Great Lakes, pp 101–128 in J. I. L. Mortsch, A. Hebb, and S. Doka, editor. Great Lakes coastal wetland communities: vulnerability to climate change and response to adaptation strategies. Environment Canada and Fisheries and Oceans Canada, Toronto, Ontario.

Dudgeon D, Arthington AH, Gessner MO, Kawabata Z-I, Knowler DJ, Lévêque C, Naiman RJ, Prieur-Richard A-H, Soto D, Stiassny MLJ, Sullivan CA (2006) Freshwater biodiversity: importance, threats, status and conservation challenges. Biol Rev 81:163–182

Earl DA, vonHoldt BM (2012) Structure harvester: a website and program for visualizing STRUCTURE output and implementing the Evanno method. Conserv Genet Resour 4:359–361. https://doi.org/10.1007/s12686-011-9548-7

Eaton DAR (2014) PyRAD: assembly of *de novo* RADseq loci for phylogenetic analyses. Bioinformatics 30:1844–1849

Eschman DF, Karrow PF (1985) Huron Basin glacial lakes: a review. In: Karrow PF, Calkin PE (eds) Quaternary evolution of the Great Lakes. Geological Association of Canada Special Paper, vol 30, pp 79–93.

Evanno G, Regnaut S, Goudet J (2005) Detecting the number of clusters of individuals using the software STRUCTURE: a simulation study. Mol Ecol 14:2611–2620

Gautier M, Gharbi K, Cezard T, Foucaud J, Kerdelhué C, Pudlo P, Cornuet JM, Estoup A (2013) The effect of RAD allele dropout on the estimation of genetic variation within and between populations. Mol Ecol 22:3165–3178

Ginson R, Walter RP, Mandrak NE, Beneteau CL, Heath DD (2015) Hierarchical analysis of genetic structure in the habitat-specialist Eastern Sand Darter (*Ammocrypta pellucida*). Ecol Evol 5:695–708

Gordon, A., Hannon, G.J. 2010. Fastx-toolkit. FASTQ/A short-reads pre-processing tools.

Hammer Ø, Harper DAT, Ryan PD (2001) PAST: paleontological statistics software package for education and data analysis v3. Palaeontol Electron 4:1–9

Jelks HL, Walsh SJ, Burkhead NM, Contreras-Balderas S, Díaz-Pardo E, Hendrickson DA, Lyons J, Mandrak NE, McCormick F, Nelson JS, Platania SP, Porter BA, Renaud CB, Schmitter-Soto JJ, Taylor EB, Warren ML Jr (2008) Conservation status of imperiled North American freshwater and diadromous fishes. Fisheries 33:372–407

Keenan K, McGinnity P, Cross TF, Crozier WW, Prodöhl PA (2013) diveRsity: an R package for the estimation and exploration of population genetics parameters and their associated errors. Methods Ecol Evol 4:782–788

Kramski N (2014) Conservation of fishes in altered ecosystems: the movement ecology of listed Grass Pickerel in an agricultural drain. M.Sc. thesis, Department of Integrative Biology, University of Guelph, Guelph, Ontario

Lujan NK, Weir JT, Noonan BP, Lovejoy NR, Mandrak NE (2020) Is Niagara Falls a barrier to gene flow in riverine fishes? A test using genome-wide SNP data from seven native species. Mol Ecol 29:1235–1249

Magurran AE, Henderson PA (2010) Temporal turnover and the maintenance of diversity in ecological assemblages. Philos Trans R Soc B 365:3611–3620

Mäkinen HS, Cano JM, Merilä J (2006) Genetic relationships among marine and freshwater populations of the European three-spined stickleback (*Gasterosteus aculeatus*) revealed by microsatellites. Mol Ecol 15:1519–1534

Mandrak NE, Crossman EJ (1992) Postglacial dispersal of freshwater fishes into Ontario. Can J Zool 70:2247–2259

Manel S, Schwartz MK, Luikart G, Taberlet P (2003) Landscape genetics: combining landscape ecology and population genetics. Trends Ecol Evol 18:189–197

McCune JL, Harrower WL, Avery-Gomm S, Brogan JM, Csergo AM, Davidson LNK, Garani A, Halpin LR, Lipsen LPJ, Lee C, Nelson JC, Prugh LR, Stinson CM, Whitney CK, Whitton J (2013) Threats to Canadian species at risk: an analysis of finalized recovery strategies. Biol Conserv 166:254–265

McNeely JA, Miller KR, Reid WV, Mittermeier RA, Werner TB (1990) Conserving the world's biological diversity, IUCN, Gland, Switzerland; WRI, CI, WWF-US, and the World Bank, Washington, DC

Nei M (1973) Analysis of gene diversity in subdivided populations. Proc Natl Acad Sci 70:3321–3323

Page LM, Burr BM (2011) Peterson field guide to freshwater fishes, 2nd edn. Houghton Mifflin Co., Boston

Palsbøll PJ, Bérube M, Allendorf FW (2006) Identification of management units using population genetic data. Trends Ecol Evol 22:11–16

Poissant J, Knight TW, Ferguson MM (2005) Nonequilibrium conditions following landscape rearrangement: the relative contribution

of past and current hydrological landscapes on the genetic structure of a stream-dwelling fish. Mol Ecol 14:1321–1331
Pritchard JK, Stephens M, Donnelly P (2000) Inference of population structure using multilocus genotype data. Genetics 155:945–959
Reed DH, Frankham R (2003) Correlation between fitness and genetic diversity. Conserv Biol 17:230–237
Rondeau EB, Minkley DR, Leong JS, Messmer AF, Jantzen JR, von Schalburg KR, Lemon C, Bird NH, Koop BF (2014) The genome and linkage map of the Northern Pike (*Esox lucius*): conserved synteny revealed between the salmonid didter group and the Neoteleostei. PLoS ONE 9:e102089
Scott WB, Crossman EJ (1998) Freshwater fishes of Canada. Galt House Publications Ltd, Oakville, Ontario
Sheldon AL (1988) Conservation of stream fishes: patterns of diversity, rarity, and risk. Cons Biol 2:149–156
Stepien CA, Murphy DJ, Strange RM (2007) Broad- to fine-scale population genetic patterning in the smallmouth bass *Micropterus dolomieu* across the Laurentian Great Lakes and beyond: an interplay of behaviour and geography. Mol Ecol 16:1605–1624
Storfer A, Murphy MA, Evans JS, Goldberg CS, Robinson S, Spear SF, Dezzani R, Delmelle E, Vierling L, Waits LP (2007) Putting the "landscape" in landscape genetics. Heredity 98:128–142
Su G, Logez M, Xu J, Tao S, Villéger S, Brosse S (2021) Human impacts on global freshwater fish biodiversity. Science 371:835–838
Sunde J, Yildirim Y, Tibblin P, Bekkevold D, Skov C, Nordahl O, Larsson P, Forsman A (2022) Drivers of neutral and adaptive differentiation in pike (*Esox lucius*) populations from contrasting environments. Mol Ecol 31:1093–1110
Ward RD, Woodwark M, Skibinski DOF (1994) A comparison of genetic diversity levels in marine, freshwater, and anadromous fishes. J Fish Biol 44:213–232
Weir BS, Cockerham CC (1984) Estimating F-statistics for the analysis of population structure. Evolution 38:1358–1370
Wright S (1942) Isolation by distance. Genetics 28:114–138
Yagi AR, Blott C (2012) Niagara River watershed fish community assessment (1997 to 2011). Ontario Ministry of Natural Resources, Fonthill, ON

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.